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Subject code & Name	:	CSE304 & Full Stack Development	Practical	:	8	Academic Year	:	**2025-26**

Task - 8

Aim

• Optimize the React frontend of the Task Management application by implementing lazy loading for route-based components.
• Apply React.lazy() and Suspense to at least the Projects and Contact routes (or equivalent route-based components).
• Provide a meaningful fallback UI while a lazy-loaded chunk is loading.
• Measure and compare bundle size and load time before and after optimization using browser developer tools.
• Document the before/after comparison with screenshots or recorded values.

Code

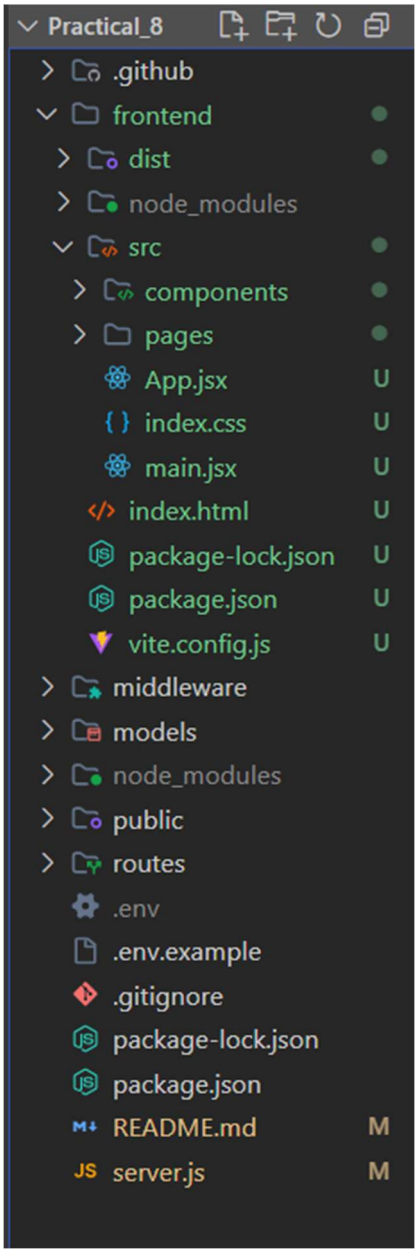
GitHub Link Repository Link:

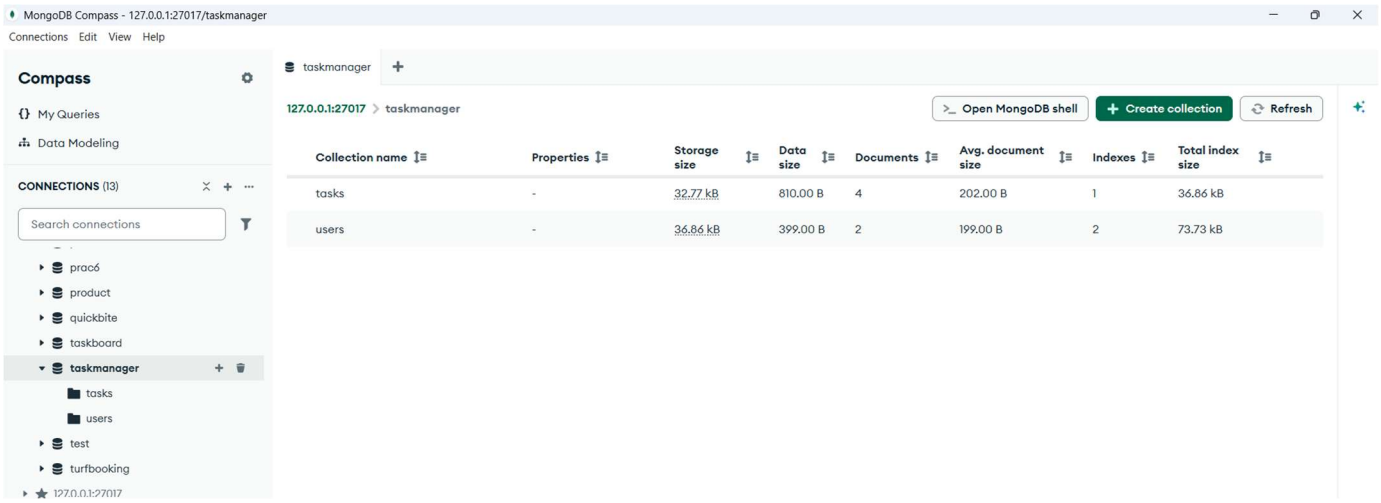
Code repo :-

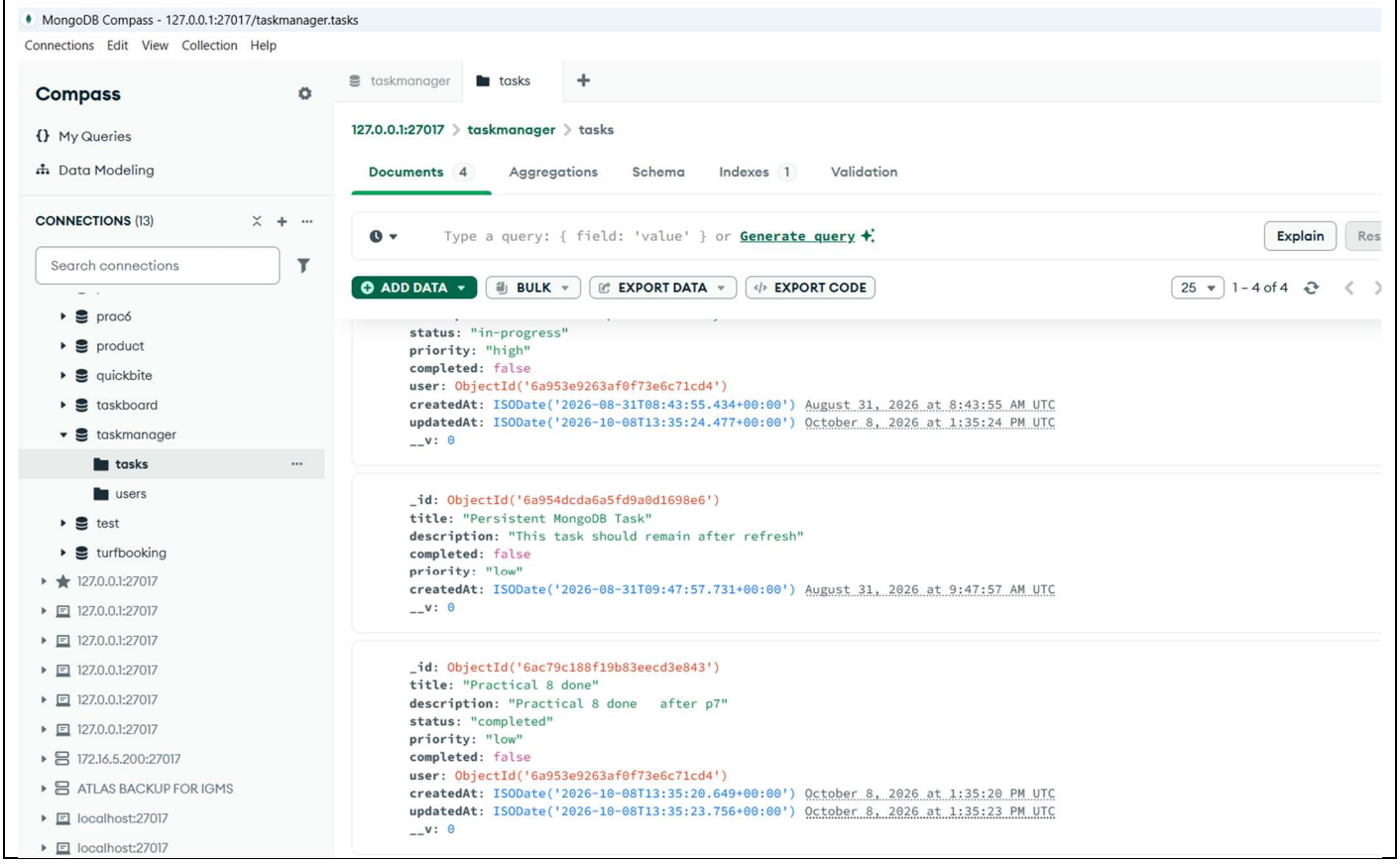
https://github.com/PRINCE-24CS072/AWDF/tree/main/Practical-8

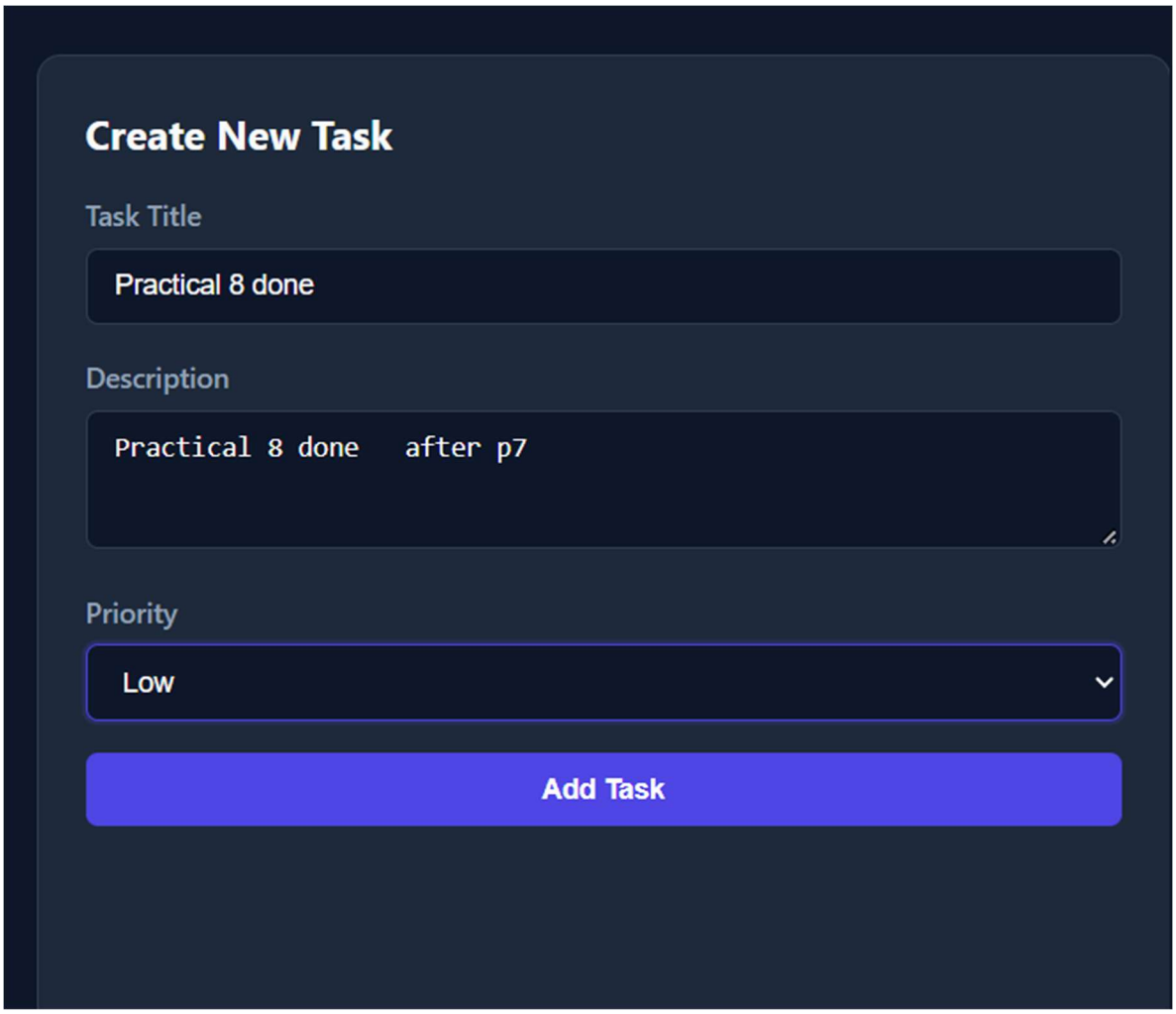
Output

Folder Structure

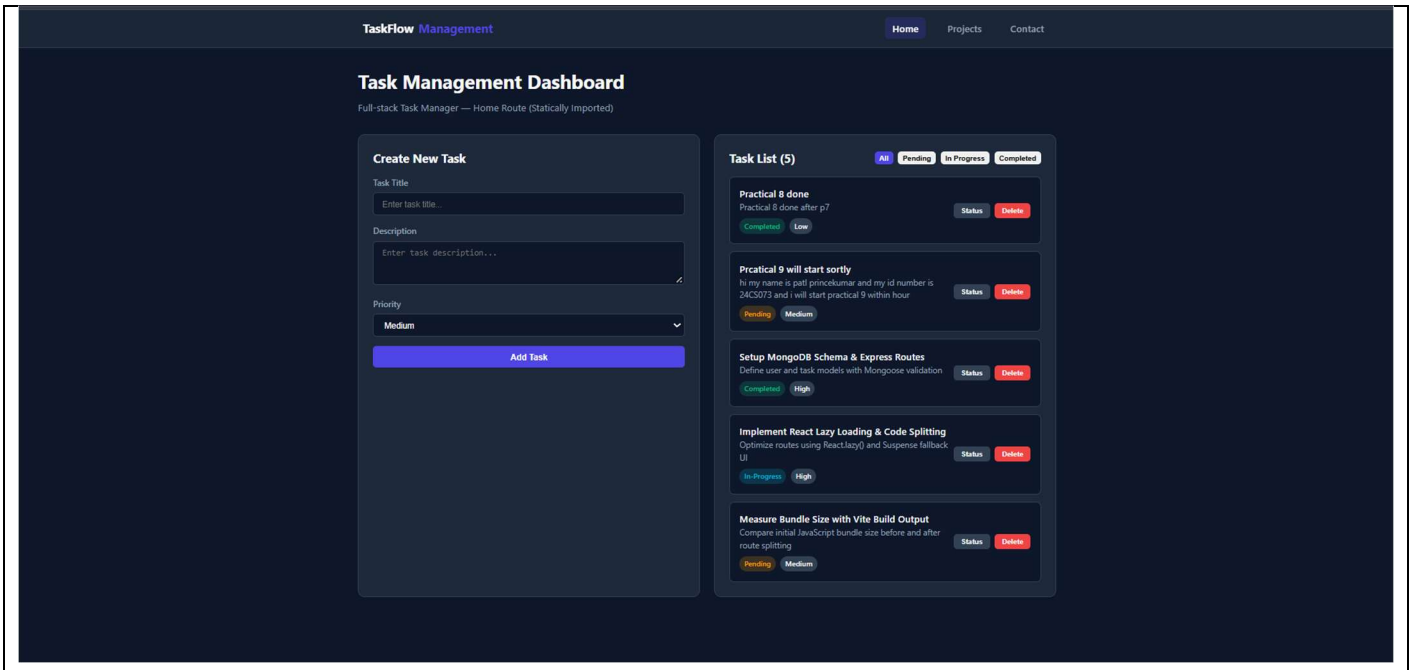


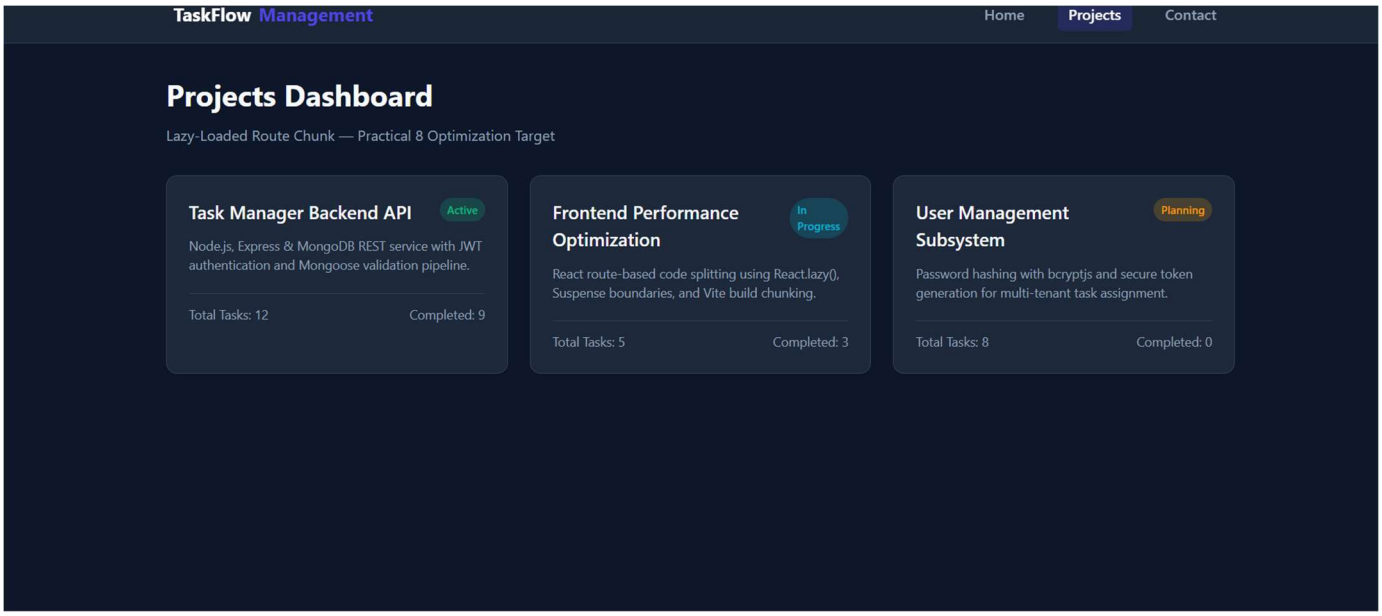

MongoDB Compass - 127.0.0.1:27017/taskmanager
Connections Edit View Help
Compass
My Queries
Data Modeling
CONNECTIONS (13)
Search connections
prac6
product
quickbite
taskboard
taskmanager
tasks
users
test
turfbooking
127.0.0.1:27017
taskmanager
127.0.0.1:27017 > taskmanager
Open MongoDB shell
Create collection
Refresh
Collection name
Properties
Storage size
Data size
Documents
Avg. document size
Indexes
Total index size
tasks - 32.77 kB 810.00 B 4 202.00 B 1 36.86 kB
users - 36.86 kB 399.00 B 2 199.00 B 2 73.73 kB

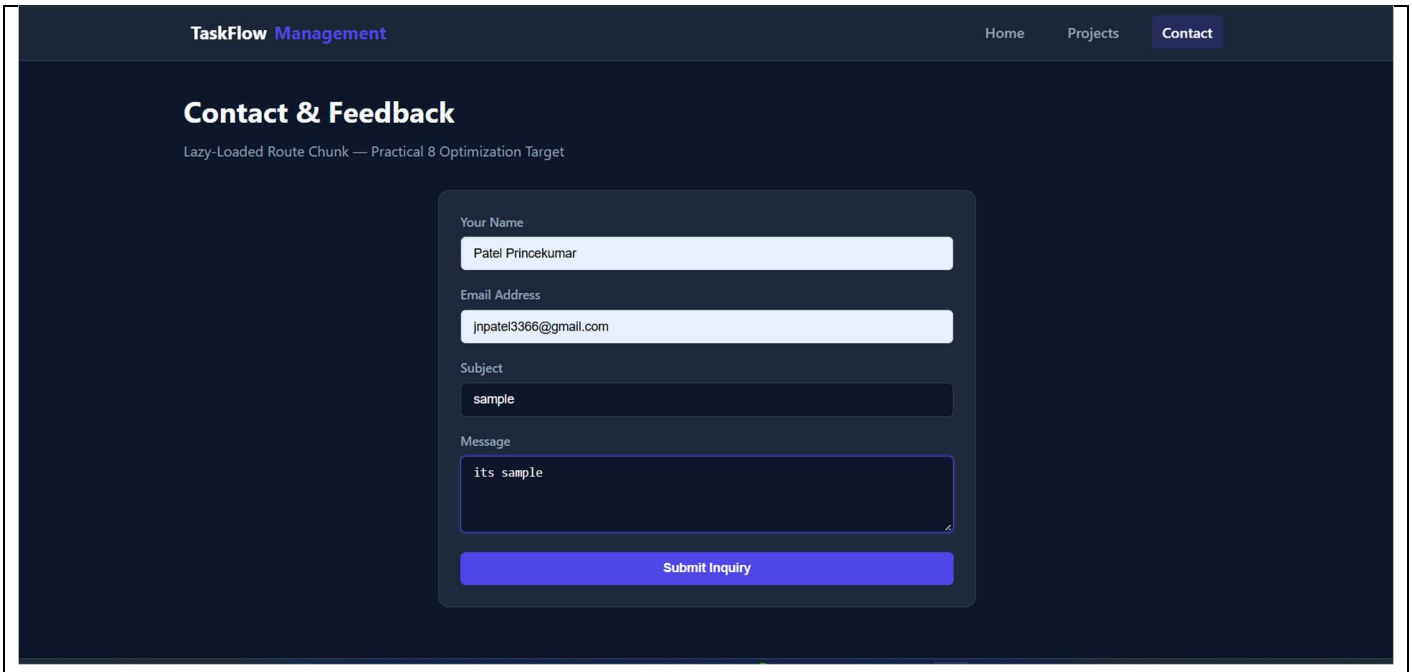

MongoDB Compass - 127.0.0.1:27017/taskmanager.tasks
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (13)
Search connections
prac6
product
quickbite
taskboard
taskmanager
tasks
users
test
turfbooking
127.0.0.1:27017
172.16.5.200:27017
ATLAS BACKUP FOR IGMS
localhost:27017
taskmanager
tasks
127.0.0.1:27017 > taskmanager > tasks
Documents 4
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 4 of 4
status: "in-progress"
priority: "high"
completed: false
user: ObjectId('6a953e9263af0f73e6c71cd4')
createdAt: ISODate('2026-08-31T08:43:55.434+00:00') August 31, 2026 at 8:43:55 AM UTC
updatedAt: ISODate('2026-10-08T13:35:24.477+00:00') October 8, 2026 at 1:35:24 PM UTC
__v: 0
_id: ObjectId('6a954dcda6a5fd9a0d1698e6')
title: "Persistent MongoDB Task"
description: "This task should remain after refresh"
completed: false
priority: "low"
createdAt: ISODate('2026-08-31T09:47:57.731+00:00') August 31, 2026 at 9:47:57 AM UTC
__v: 0
_id: ObjectId('6ac79c188f19b83eecd3e843')
title: "Practical 8 done"
description: "Practical 8 done after p7"
status: "completed"
priority: "low"
completed: false
user: ObjectId('6a953e9263af0f73e6c71cd4')
createdAt: ISODate('2026-10-08T13:35:20.649+00:00') October 8, 2026 at 1:35:20 PM UTC
updatedAt: ISODate('2026-10-08T13:35:23.756+00:00') October 8, 2026 at 1:35:23 PM UTC
__v: 0




```
_id: ObjectId('6ac79c188f19b83eecd3e843')
title: "Practical 8 done"
description: "Practical 8 done   after p7"
status: "completed"
priority: "low"
completed: false
user: ObjectId('6a953e9263af0f73e6c71cd4')
createdAt: ISODate('2026-10-08T13:35:20.649+00:00') October 8, 2026 at 1:35:20 PM UTC
updatedAt: ISODate('2026-10-08T13:35:23.756+00:00') October 8, 2026 at 1:35:23 PM UTC
__v: 0
```


TaskFlow Management
Home
Projects
Contact
Task Management Dashboard
Full-stack Task Manager — Home Route (Statically Imported)
Create New Task
Task Title
Enter task title...
Description
Enter task description...
Priority
Medium
Add Task
Task List (5)
All
Pending
In Progress
Completed
Practical 8 done
Practical 8 done after p7
Completed
Low
Status
Delete
Pcratical 9 will start sortly
hi my name is patl princekumar and my id number is 24CS073 and i will start practical 9 within hour
Pending
Medium
Setup MongoDB Schema & Express Routes
Define user and task models with Mongoose validation
Completed
High
Implement React Lazy Loading & Code Splitting
Optimize routes using React.lazy() and Suspense fallback UI
In-Progress
High
Measure Bundle Size with Vite Build Output
Compare initial JavaScript bundle size before and after route splitting
Pending
Medium


TaskFlow Management
Home
Projects
Contact
Projects Dashboard
Lazy-Loaded Route Chunk — Practical 8 Optimization Target
Task Manager Backend API
Active
Node.js, Express & MongoDB REST service with JWT authentication and Mongoose validation pipeline.
Total Tasks: 12
Completed: 9
Frontend Performance Optimization
In Progress
React route-based code splitting using React.lazy(), Suspense boundaries, and Vite build chunking.
Total Tasks: 5
Completed: 3
User Management Subsystem
Planning
Password hashing with bcryptjs and secure token generation for multi-tenant task assignment.
Total Tasks: 8
Completed: 0



Terminal

```
> practical-7-task-management@1.0.0 dev
> nodemon server.js

[nodemon] 3.1.14
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
MongoDB connected successfully
Server running on http://localhost:5000
GET / - 2026-08-31T08:41:28.256Z
GET /css/style.css - 2026-08-31T08:41:28.300Z
GET /js/app.js - 2026-08-31T08:41:28.303Z
GET /favicon.ico - 2026-08-31T08:41:28.763Z
```